Statistical anomaly detectors

AI service: risk (Isolation Forest and LOF) · Run 09 Nov 2026, 01:01 UTC · **All targets met**

What is measured

Whether the two unsupervised detectors flag claims that are statistically unusual without flagging ordinary ones. They are fitted on unlabeled synthetic claims; labels exist only in this evaluation set (D-083).

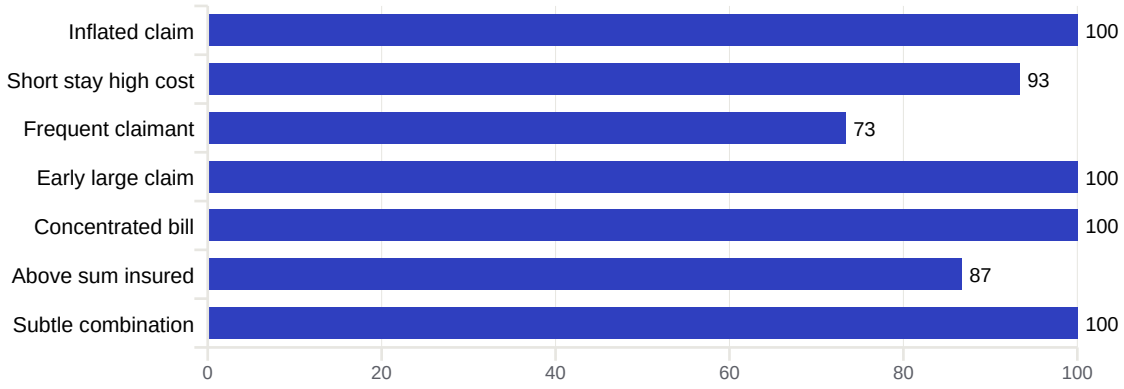
Data

Fitted on 5,000 synthetic baseline claims. Evaluated on 2,000 held-out normal claims and 210 controlled anomalies of 7 types. Seeds are fixed, so every run gives the same numbers.

Results

Measure	Result	Target	
False positives, Isolation Forest	**0.9%**	≤ 2%	**Met**
False positives, LOF	**1.1%**	≤ 2%	**Met**
Recall, either detector	**94.8%**	≥ 90%	**Met**
Recall, Isolation Forest	**47.6%**		
Recall, LOF	**93.3%**		
ROC AUC, Isolation Forest	**0.902**		
ROC AUC, LOF	**0.995**		
False positives, either detector	**1.8%**		

Recall by anomaly type, LOF (%)



Recall by anomaly type

Anomaly type	Isolation Forest	LOF
Inflated claim	0%	100%
Short stay high cost	27%	93%
Frequent claimant	17%	73%
Early large claim	87%	100%
Concentrated bill	13%	100%
Above sum insured	93%	87%
Subtle combination	97%	100%

Observations

• Isolation Forest misses inflated claims almost entirely: nearly every baseline claim asks for exactly its bill, so its random splits rarely isolate that feature. Rule FIN-03 catches inflated claims deterministically, which is why rules and detectors are combined.
• A flagged detector adds 10 points, and both together at most 20, below the HIGH threshold of 45. Statistics alone can never make a claim high risk.

Limitations

• The detectors learned the generator's idea of normal and are tested on its idea of anomalous. This shows the implementation works as designed. It says nothing about detecting fraud in real claims, which needs real data and the privacy gate first.

Reproduce

```
cd services/ai; python -m uv run python evals/train_risk_models.py
```